NORTH-EASTERN RAILWAY.

Board of Trade, (Railway Department,)
8, Richmond Terrace, Whitehall, London, S.W.,
9th December 1892.

SIR,

I HAVE the honour to report, for the information of the Board of Trade, in compliance with the instructions contained in the Order of the 2nd ultimo, the result of my enquiry into the circumstances attending a fatal collision which took place on the 2nd ultimo at Manor House signal-cabin, between Otterington and Thirsk stations, on the North-Eastern Railway.

In this case the second portion of the up Scotch express, which left Edinburgh at 11.1 p.m. on the 1st ultimo, Newcastle at 2.38 a.m., and Darlington at 3.39 a.m., when running at full speed between Otterington and Thirsk, came into collision at 4.2 a.m. or 4.3 a.m. with the rear of the 2 a.m. up goods train from Middlesboro' to Starbeck, which had passed Northallerton at 3.38 a.m., and Otterington at 3.47 a.m., and was moving slowly forward, having just started from the up home-signal at Manor House signal-cabin, where it had been kept standing for several minutes.

The goods train consisted of engine and tender, 34 waggons, all but two or three loaded, and rear brake-van; and the express was made up as follows, viz.:—engine and tender, Great-Northern brake-van, East Coast third-class carriage, Pullman car, East Coast composite carriage, brake-van, sleeping carriage, third-class, composite, and third-class carriages, Great-Northern saloon, Great North of Scotland horse-box, Caledonian van, and East Coast brake-van, or 13 vehicles (equal in weight to 15), of which 10 were fitted with the Westinghouse-brake, and the other three with through pipes.

The collision was one of great violence and the result was most disastrous. The brake-van of the goods train and one of the goods waggons were broken to pieces, and two others left the rails; the engine and tender of the express were thrown off the rails and upset on the left-hand side across a siding, lying with their wheels together nearly at right angles to the main line, and fortunately clear of it; the two next vehicles were practically demolished by the body of the Pullman car, which was shot off the bogie frames, and landed partly on the siding and partly on the main line, south of the place where the engine and tender, and the wreckage of the goods van and the two leading vehicles of the express were lying; partly on the wreckage, and on the engine, was the leading composite carriage, and underneath the trailing end of it, and standing foul of the down main line was the sleeping carriage. On the top of these two last-named vehicles was the brake-van, the fifth vehicle in the train, with its leading end telescoped into the top of the composite carriage, and with the leading end of the next third-class carriage driven underneath it; the eighth vehicle, a composite carriage, was telescoped into the third-class carriage in front of it; the ninth, a third-class carriage, and the tenth, a saloon, were off the rails, but the three rear vehicles did not leave the rails.

Nine passengers, all of whom are believed to have been in the leading third-class carriage, were either killed upon the spot or so badly injured that they died within a very short time, and 39 others were injured, some of them seriously. The guard of the goods train was killed, the driver and fireman of the express train were injured, the former very seriously; the conductor, who had fortunately changed at Darlington from the front van to the centre van, was badly shaken, and the guard who was in the rear van was slightly shaken.

The damage to the rolling-stock, of which a full return is appended, was very great, the wreckage of the two leading vehicles, and the 4th, 5th, 6th, and 7th vehicles, and to some extent the 8th vehicle, of the express, and the brake-van of the goods train, with eight of the goods waggons, having been consumed by a fire which broke out near the engine about an hour after the accident.

In the permanent-way 105 yards of the up line were broken up, and the following materials have been used to make good the damage:—210 yards of rails, 142 chairs, 106 cushions, 176 keys, 28 fish-plates, 62 fish-plate bolts, 386 spikes, and 112 sleepers.

This report has been delayed in consequence of the serious injuries to the driver of the express, whose evidence I was unable to take until the 6th instant.

Description.

Manor House block signal-cabin, the scene of this disaster, is 1 mile 30 chains south of Otterington station signal-cabin, 4 miles 73 chains south of Northallerton,

1 mile 18 chains north of Avenue junction signal-cabin, and about 3 miles north of Thirsk station.

It is a small cabin on the east side of the main line, which runs from Otterington quite straight from north to south, on a falling gradient of 1 in 628, and there is only one set of points for a short down siding.

There are distant, home, and starting signals in each direction, and the cabin contains a frame with seven working and three spare levers, four block telegraph instruments, and an ordinary needle instrument for communicating messages.

The up home-signal is 57 yards north of the cabin, on the west side of the line, being upon the same post as the down home-signal, and the actual point of collision was, as near as can be estimated, 35 yards north of the signal, the point where the engine lay after the collision being about 10 yards north of the signal.

The up and down starting-signals are 351 yards south and north of the cabin, and the up distant-signal is 849 yards north of the cabin.

The floor of the cabin is 9 ft. 6 in. above the level of the rails, and the lamp and arm of the up home-signal are respectively 24 ft. 8 in. and 34 ft. 9 in. above this level.

The line is worked upon the absolute block telegraph system under North-Eastern Railway Company's Rules, from which the following are extracts:—

"2. When commencing to signal a train, the 'Call attention' signal must be the first signal given; all signals must be acknowledged by repeating them, and no signal must be considered as understood until it has been correctly repeated to the sending station."

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"4. For the purpose of illustrating the course to be adopted, let A, B, and C represent three block signal-cabins, and the process of signalling a train is as follows:—On the approach of a train to cabin A the signalman will 'Call attention' of the signalman at cabin B, and then give the signal 'Be ready,' and the signalman at cabin B, after having, in accordance with Rule 7, ascertained that the line is clear upon which the train is to run, must repeat the signal and peg the needle to 'Line clear.' As soon as the train has passed cabin A the signalman there must give the bell signal 'Train on line,' upon which the signalman at cabin B must acknowledge the signal and unpeg the needle. The signalman at cabin A must then indicate to the signalman at cabin B the kind of train that is approaching by giving the proper dial signal, which the signalman at cabin B must acknowledge by repeating, and having received the intimation that his acknowledgment is correct, by the signalman at cabin A holding the needle over to 'Train on line,' he must peg it in that position, and then 'Call attention' of and give the signal 'Be ready' to the signalman at cabin C. As soon as the train has passed cabin B it must be signalled in a similar manner to the signalman at cabin C, who must forward the signal 'Be ready' to the next cabin, and so on throughout the block. When the train has *passed* the advance signal at cabin B, and is continuing its journey in the next section, or has been shunted clear of the main line, the signalman at cabin B must 'Call attention' of the signalman at cabin A and give the proper signal, indicating that the line is clear of the train, which must be duly acknowledged by the signalman at cabin A."

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"5. No train must be allowed to pass or start from any station or junction where the block system is in operation, nor any obstruction of the line be allowed to take place, without the signals having been previously made and acknowledged as herein directed."

"7 (*a*). In dealing with trains of every description, except as provided for in clauses (*b*), (*d*), and (*e*), the line in the rear section must always be kept blocked until the preceding train has either been shunted clear of the main line or has passed the advance-signal, and is continuing its journey in the next section, or, where there is no advance-signal, has passed the home-signal at least 400 yards, or the cabin in advance if within this distance, and is continuing its journey in the next section."

The following are extracts from the Company's book of rules and regulations:—

"90. The signal-boxes at which the block telegraph working is in operation are furnished with instruments to signal for each line of rails, and the system under which these instruments are to be worked, and the mode of indicating the description of approaching trains, will be laid down in the code of regulations supplied to signalmen or exhibited in the signal-boxes for the guidance of the persons in charge."

"91. On those portions of the line worked on the absolute block system a second train or engine must not be allowed to enter a section until the preceding train or engine has been signalled as having passed out of the section, except under the circumstances specified in Rules 100 and 101, to meet cases of train or telegraph failure."

* * * * *

"275A. * * * * * *
In case of detention at a home or starting signal the engine-driver must sound his whistle, and if still detained the guard or fireman must go to the signal-box and remind the signalman of the position of the train or engine, and remain there until the signalman can give permission to go forward. In foggy weather or during falling snow the guard or fireman must, immediately upon the train or engine coming to a stand, proceed to the signal-box.

"The duty must be performed by the guard or fireman in accordance with the following instructions:—

* * * * *

"In the case of a goods train with only one guard, when stopped at the home-signal, by the fireman.

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"281. Should an accident to a train foul or be dangerously near to any line of rails used by trains running in the opposite direction, the engine-driver of the disabled train must immediately detach his engine (if it be able to run forward) and proceed with it 1,200 yards from the scene of the accident, and leave his fireman with detonators to act as laid down in Rules 213 and 214 to protect the opposite line of rails. The engine-driver must then go forward to the nearest signal-box and inform the signalman of the obstruction, in order that any train or engine running on the opposite line may be stopped." * * * *

"213. When a train is stopped by an accident or from any cause (except where it is efficiently protected by fixed signals), the guard, if there be only one, or the rear guard, if there be more than one, must immediately go back 1,200 yards to stop any following train." * * * *

Evidence.

George Birch states: I have been about 24 years in the service, and a signalman for about 20 years; at the Northallerton north junction for nearly ten years. On the 1st November I came on duty at 10 p.m. for eight hours, having been off duty for 16 hours. I was informed by telegraph from Newcastle, Ferry Hill, and Darlington, how the second portion of the Scotch express was running. There was above 20 minutes margin between the two portions all the way. I cannot remember the exact interval. The following are the block-signals I received for the two portions of the express, and the Middlesboro' goods train:—

First portion of Express.

"Be ready" received from Castle Hill junction	3.31
Accepted at once.	
"On line" received	3.32
Train passed	3.33
"Line clear" given to Castle Hill junction	3.33
"On line" given to Hutton Wood	3.33
"Line clear" received from Hutton Wood	3.36

Middlesboro' Goods Train.

"Be ready" received from Northallerton low junction	3.20
Signal not accepted until	3.36
"On line" received from Northallerton low junction	3.36
Train passed	3.38
"Line clear" given to Northallerton low junction	3.38
"On line" given to Hutton Wood	3.38
"Line clear" received from Hutton Wood	3.44

Second portion of Express.

"Be ready" received from Castle Hill junction	3.55
Accepted at once.	
"On line" received	3.55
Train passed	3.58
"Line clear" given to Castle Hill junction	3.58
"On line" given to Hutton Wood	3.58
"Line clear" received from Hutton Wood	4.0

At the time I accepted the goods train from the low junction the second portion of the express had not left Darlington. It left Darlington at 3.39, as I was informed by telegraph. The margin which is ordered to be kept at my cabin in the case of ordinary goods trains in front of fast passenger trains is 20 minutes. There were three tail-lights on the goods train burning properly, one light in the centre of the buffer-beam, and two at the sides of the van higher up. It was sometimes rather foggy that night, but not very bad at Northallerton. I could see signal lights over 100 yards off. I could see my up starting-signal light.

2. *Edmund Wright* states: I have been 16 years in the service, and a signalman for seven years; at Hutton Wood for the whole time. On the 1st November I came on duty at 6 p.m. to work till 6 a.m. I had left duty at 6 a.m. There are five working levers in my cabin; four block-instruments, and a speaking instrument. I believe my clock was right. The following are the block-signals recorded

in my book for the two portions of the Scotch express, and the Middlesboro' goods train:—

First portion of Express.

"Be ready" received from Northallerton high junction	3.30	a.m.
"On line" received from Northallerton high junction	3.33	"
Train passed	3.35	"
"Line clear" given back	3.35	"
"On line" given to Otterington	3.35	"
"Line clear" received from Otterington	3.37	"

Goods Train.

Be ready" received from Northallerton north junction	3.33	"
Accepted at once.		
'On line"	3.37	"
Train passed	3.43	"
"Line clear" given back	3.43	"
"On line" given to Otterington	3.43	"
"Line clear" received from Otterington	3.47	"

Second portion of Express.

"Be ready" received from Northallerton north junction	3.55	"
Accepted at once.		
"On line" received	3.57	"
Train passed	3.59	"
"Line clear" given back	3.59	"
"On line" given to Otterington	3.59	"
"Line clear" received from Otterington	4.1	"

It was a foggy night. It was very variable. I did not think it necessary to send for fogmen. I could see my advance-signal light at times, but at times not. I did not notice it was thick till after midnight. A down goods train passed me at 3.33, and another at 3.43. That was the last until 7.15.

3. *Henry Eden* states: I have been 21½ years in the service, and a signalman for about 8½ years; at Otterington since August 1891. On the 1st November I came on duty at 7.15 p.m. to work till 7.25 a.m. The proper hours are from 6 p.m. to 6 a.m., but owing to my mate being on a holiday, and the relief man not being able to get to Otterington earlier, there was a temporary alteration in the hours. I had left duty at 1.25 p.m., on Monday, having been on duty since midnight on Sunday. I should have been relieved at 6 a.m., only there was some mistake about the relief man coming. There are nine working levers, and one spare lever, four block instruments, and a speaking instrument in my cabin. We work absolute block under the ordinary rules. I produce the train record book for my cabin. The following are the signals recorded for the two portions of the Scotch express, and the Middlesboro' and Starbeck goods:—

First Portion of Express.

"Be ready" received from Hutton Wood	3.33	a.m.
Accepted at once.		
"On line" received from Hutton Wood	3.35	"
Train passed	3.37	"
"Line clear" given to Hutton Wood	3.37	"
"On line" given to Manor House	3.37	"
"Line clear" received from Manor House	3.38	"

Goods Train.

"Be ready" received from Hutton Wood	3 38	a.m.
Accepted at once.		
"On line" received from Hutton Wood	3.43	"
Train passed	3.47	"
"Line clear" given to Hutton Wood	3.47	"
"On line" given to Manor House	3.47	"
Line clear" received from Manor House	4.0	"

The last entry is meant for 4.0. The figure 3 in the hour column was put down in anticipation, and, as the signal was not received until 4, I put the 4, and what I mean for a 0 in the minute column.

Second Portion of the Express.

"Be ready" received from Hutton Wood	3.58	a.m.
Accepted at once.		
"On line" received from Hutton Wood	4.0	"
Train passed	4.1	"
"Line clear" given to Hutton Wood	4.1	"
"On line" given to Manor House	4.1	"
"Line clear" received from Manor House	4.3	"

For the three last entries I made the 4 in anticipation, and as I did not enter them until some time after the accident, after I had fetched the platelayers, I made a mistake in the minute column and put 41 instead of 1. I also entered the "Line clear" for the express at 4.3, which is the time it should have been cleared, but it never was cleared. I was rather confused. After I had given the train "On line" to Manor House, but before I entered it, I went to the window to look after it, and then, when going back to book, was called on the speaking instrument from Manor House. The signalman asked me if it was the E.P. that I had put "On line." I answered "Yes." I thought it was strange, and I went to the window wondering what was up with Holmes in asking that question, and I heard the pitch in. It was a violent crash. Holmes called up again on his speaking instrument, and offered me a D.B. message. I ran to the block instrument, and asked him what was amiss. He said, "Tell Mr. Kirby that the express has run into a goods." I told the signalman at Hutton Wood to caution up trains to stop at my cabin, then went to Mr. Kirby, and then went and called the platelayers. It was a thick night but variable. It was too thick for me to see the Manor House up distant-signal light. Some time during the night Holmes had discussed with me whether it was necessary to have out the fogmen, but it got lighter and we did not do so. At the time the accident happened I did not think it was so thick as to require fogmen. I could see the signal lights on my advance-signal 350 yards off. I could see the back light of my home-signal, but not of my distant. I could see the tail light on a passing train. The first portion of the express had two tail lights which indicated to me that there was a second portion to follow. I had no other intimation that the train was running in two partions that night. Holmes had come to my cabin before going to his own cabin at about half-past 7 p.m. He asked me to wire to him if his mother arrived by the later train, and then said, "Harry, I am about done to start duty." He said, "I have never been off my legs since 12 o'clock. "I have had to walk from Northallerton and twice "to Otterington and back." Holmes' mother did arrive by the train at 8.58 p.m., and I wired to Holmes to tell him so. When Holmes was in my cabin he looked tired and upset. Holmes, when he went on duty, said he was done up, but he did not tell me actually that he was unfit for duty. He looked put out. He was just outside my box. I thought he was most troubled about his wife, and it did not strike me that there was any risk on his going to duty. He worked as well as possible up to the time that the second portion of the express passed, taking all the signals at once. A goods train usually takes from three to four minutes in running from my box to Manor House. I did not notice that I had not received the "Line clear" signal for the goods train until the second portion of the express had been offered to me. I was sitting down by the fire reading. If I had noticed that the goods train had been so long away I would probably have rung up to ask about it. According to our block regulations it is right to offer the "Be ready" for a

second train before the train out of section-signal, or the "Line clear" signal as we call it, for the preceding train has been received. I offered the "Be ready" for the second portion of the express at 4 p.m. I got back the attention-signal, then the "Line clear" signal, and then the acceptance of the "Be ready," and I lowered my signals for the train. All the block-signals were given in a proper manner and acknowledged at once. I had just gone across the cabin to book the train when I got the message from Holmes to ask whether the train was the E.P. Holmes offered me the message on the speaking instrument, but I took it on the block instrument. The reason I did so was that I knew something was amiss, and thought it was the quickest way to find out. Manor House is closed on Sundays. I think my clock was correct that night. I think that about 90 to 100 trains pass my box on the up and down lines.

4. *James Holmes* states: I have been about 11 years in the service, and a signalman for the whole time; at Manor House for 15 months. On the 1st November I came on duty at 8 p.m. to work till 6 a.m. on the 2nd November. I had gone off duty at 6 a.m. on the 1st November after 12 hours' work. I had arranged with my mate to work until 8 p.m. on the 1st November, with the consent of the station-master at Otterington. There are seven working and three spare levers in my cabin, four block instruments, and a speaking instrument. We work absolute block in both directions under the ordinary rules. I believe the clock in my cabin was correct. It is, as a rule, corrected every morning at 10 a.m. I produce the train record book for my cabin. The following are the block-signals recorded for the first portion of the up Scotch express:—

"Be ready" received from Otterington at - - - - -	3.35 a.m
Accepted at once.	
"On line" received from Otterington at - - - - -	3.37 "
Train passed - - - - -	3.38 "
"Line clear" given back - -	3.38 "
"On line" sent to Avenue junction -	3.38 "
"Line clear" from Avenue junction not recorded.	

The "Be ready" would have been sent forward as soon as I received the "Train on line" from Otterington. The "On line" signal sent to Avenue junction at 3.38 a.m. is the last signal recorded in the book. I saw the train passing and it had a double tail-light on. That intimated to me that there was a second portion running. That was the only intimation I had. At the busy season of the year this train often runs in two portions. I remember soon after the train passed getting the "Be ready" for an up goods train in the usual way, and I accepted it. The signals had been put to danger behind the first portion of the Scotch express, and I did not lower them for the goods train. I did not signal forward the goods train. After that I must have been overpowered by sleep, and I do not remember the arrival of the goods train, or anything else, until I got the "Attention" signal and the "Be ready" from Otterington for the second portion of the express. I only got it once, I believe, and I gave the "Line clear" signal back, thinking I had omitted to give it for the first portion of the Scotch express, and then accepted the "Be ready." I then lowered my signals, home, distant, and starting. The goods train, which had been standing at my home-signal took this signal and started. I heard no whistle from this train either before or after it started. For a moment I seemed powerless to do anything, but as soon as I could collect myself I asked the signalman at Otterington, on the block instrument, "Is this the second portion "of the express?" He replied, "Yes." I had thrown my home and distant signals to danger just before I asked this question, but it was too late. I would like to say something in explanation for the state I was in. On Monday 31st October, not having been on duty on the previous day, I got up at about 9 a.m., and I worked in my orchard pulling apples for the market, and did not go to-bed during the day, as I usually do when going upon night duty. I went on duty at 6 p.m., the usual time, and did not feel at all fatigued, but went through my duty in the usual way. I came off duty at 6 a.m., and got home at about 7.15 a.m. I live about 2½ miles from the cabin, and when on night duty I usually wait for the Avenue junction signalman, who lives near me, and we walk home together. On reaching home my wife had not got the fire lighted, so I lighted it myself. While I was doing this my wife said to me, "I do not think "our Rosy is very well this morning," and that "she seemed to be breathing rather hard." I said, "Probably she has a little cold, leave her in bed with "me, and I will go to-bed and have a good day's "rest." I said, "Get Teddy off to school and do "what is necessary for the other children, and then "you may get me a little breakfast if you like." She brought me a cup of tea, and something to eat, at about quarter-past 9, I think. I then looked at the child. She seemed to be sleeping all right, and I said she might leave her a little longer. She came again about a quarter to 12 to fetch down the child to wash and dress her, and said that as she had not heard it cry or anything she thought it was all right. She took the child away and did not disturb me in the least when doing so. The next I remember was my wife calling out "Jim, Jim, I believe this child is in a fit." I jumped out of bed at once to see what was the matter, and told my wife to run for a neighbour who had more experience. She came, and told me to go for a doctor, which I did. I first went to Otterington station, about a mile, and got a lift on a Darlington goods train to Northallerton. When I got to the doctor's house the servant told me that he was out. She asked my name and where I resided. On hearing from me where I lived she said that the doctor had gone round by Enderby, and would then go round by Otterington to Thornton-le-Moor, where I live. I said, "Then they would probably call him in "if they saw him passing, but that at any rate I "would walk home so that I might perhaps meet "him." The distance is about 4½ miles. I did not meet him, but he telegraphed later to ask me if I had got another doctor, and I answered that when I got home I found the child was dead. It was about 12 o'clock when I started from home, and I got home at about 2 p.m. After I got home and heard that the child had died, my wife said she really could not stop alone that night, and I said I would telegraph for my mother, but that really I was not fit for duty that night, and would try to get relief. I had to pass the station on my way to the telegraph office, and I went to see Mr. Kirby, the station-master at Otterington, under whose orders I work. I said to him, "I have "had a child die very suddenly. Will you wire to "Mr. Pick (the signal inspector) to see if I can be "relieved for the night?" as I knew that I was not fit for duty. He said he would try what he could, and that I had better come up at the usual time for duty, and see what the reply was about the relief. I went about 6 p.m. to the station expecting to meet my mother on arrival from York, and when I got there I asked Mr. Kirby what the reply was. He said that Mr. Pick had replied that he regretted he could not relieve me that night. On hearing this I went into Otterington cabin, and wired to my mate at Manor House, telling him the circumstances of the case, and asking him if he could stop on a short time. He said I was not to put myself out, or to hurry myself. I replied I would come as soon as I could. I went home and told my wife that my mother had not come, and that she had better get a neighbour to be with her during the night, that I would take what refreshment I wanted, and would then go to the station to see if my mother had come by the slow train, and would then go on duty. My mother did not come by that train and I asked the Otterington signalman

to wire to me if she came by a later train. I said I would then go to duty although I was not fit for it. I arrived at the cabin at about 8 p.m. I did my work quite well until the first portion of the express passed and was priding myself on keeping up, and then I seemed to become confused all at once. I did not feel inclined to eat before going to the cabin, but after I got there I did eat something. All the drink I had had during the day, except tea, was a glass of beer in Northallerton when on my way home at about 1 p.m. It was very thick at the time of the accident. If it had not been I would have seen the engine of the goods train standing at my home-signal when I looked out before lowering the signal for the second portion of the express. As far as I can recollect, I sent the "Be ready" signal for the second part of the express to Avenue junction as soon as I received it from Otterington. It is the rule to forward this signal as soon as received for up express passenger trains, on account of the short section from Otterington to Manor House. The "Be ready" was accepted by Avenue junction at once, and before I lowered my signals for the express.

5. *Thomas Gibson* states: I have been between 15 and 16 years in the service as a signalman, and at Avenue junction for 13 years. On the 1st November I came on duty at 20 minutes to 6, to work till 6 a.m. I came voluntarily on a little before my proper time, which is 6 p.m. I had been off duty for 12 hours. There are in my cabin 13 working levers, four block instruments, speaking instruments, and telephone to Thirsk. We work absolute block system on the main line. The following are extracts from my train record book for the first portion of the Scotch express :—

"Be ready" received from Manor House - - - -	3·36 a.m.
Accepted at once.	
"On line" received from Manor House	3.39 „
Train passed - - - -	3.40 „
"Line clear" given to Manor House	3.40 „
"On line" given to Thirsk north junction - - - -	3.40 „
"Line clear" received from Thirsk north junction - - -	3.41 „

My clock was right. I got no signals at all for the goods train. The only signal I got for the second portion of the Scotch express was the "Be ready" at 4 a.m., which I accepted at once, pegging my needle over to "Line clear." The Scotch express was running late, being due to pass Thirsk at 3 a.m. The goods train was, I believe, running at about right time. If there is sufficient margin between the two portions of the Scotch express, it is usual to let the goods come forward, as it runs into the independent line at my cabin. There are no sidings long enough for a long goods train at Manor House or Otterington. It was rather foggy at times that night, but I could see my advance-signal lights. I did not think it was so thick that there was a necessity for fogmen to be out. The last down train passed my box at 3.32 a.m., and I received "Line clear" for it from Manor House at 3.36 a.m.

6. *Thomas Kirby* states: I have been about 26 years in the service, and station-master at Otterington for 4½ years. On the 1st November, signalman Holmes, who was on night duty at Manor House, came to me shortly after 2 p.m. and said, "Mr. "Kirby, I want you to telegraph to Mr. Pick to ask "him if he can relieve me to-night." I saw he looked out of sorts, and I said, "What's amiss Holmes?" He said, "My baby has died suddenly in a fit." I said, "Has it been ill long?" and he said that it had been a little amiss, but they had not thought much about it. He gave more particulars about the illness. On that morning, he said, his wife had left the baby in bed with him, and had found it in a fit about 12 o'clock. I told him I would telegraph for a relief, but I was afraid they might be all engaged with such short notice. I telegraphed to Mr. Pick, who is the signal inspector, "Can you "send relief to Manor House to-night? Holmes' "child dead." Holmes had told me that he felt in "bad fettle" for duty; but I did not wire to Mr. Pick to that effect, but only that Holmes' child was dead. Holmes said he would be down again at about 6 p.m., and that perhaps I could have a reply by then. He came shortly before 6, and I told him the reply I had got—which was, "Cannot relieve Holmes to-night." He said, "It's a bad job. I shouldn't care so much "for myself, but I am most bothered about my wife, "as she is not fit to be left all night by herself. I "have sent for my mother, and I will wait and see if "she comes by this train, and if she doesn't, I'll go "back and get some of the neighbours to stay with "my wife, and then come and meet the half-past 7 "train, and go from it to duty; but I'll ask Creed "(his mate) to remain on duty till 8 p.m. to allow "me to do this." I agreed to this as I thought it would assist Holmes. He was distressed about his child, but I did not think he looked unfit for duty. I got the answer from Mr. Pick at about 4.40 p.m. Holmes has been steady since he has been with me. I did not see him after the train arrived at 7.37, but his mother did not come by that train. I looked after her when she arrived at 8.58, and gave orders for signalman Eden to let Holmes know she had arrived and had gone to his house. On the first occasion when Holmes came to me at 2 p.m., he said to me, after asking me to telegraph to Mr. Pick, that "he did not feel fit for duty," and then I asked him what was amiss with him. It was on the second occasion that he came at 20 minutes to 6 that he said he felt in "bad fettle." My staff at Otterington station consists of two clerks and two porters, and two signalmen working alternately at the cabin. There was no one I could have sent qualified to take Holmes' duty. At 6 p.m. there was one clerk and two porters on duty, and one porter had to leave immediately, as he had been on duty since 6 a.m. If I had thought it necessary to switch out Manor House, I would have been justified in taking the responsibility of doing so. If I had thought Holmes was incapable of doing his duty I would have done so. I thought from his manner that he was more troubled about his family than about himself, and I took particular care that he should be informed that his mother had arrived so as to relieve his mind. I got to the scene of the accident at about half-past 4. I went on the engine of an up goods train which had followed the second portion of the express, and had arrived at Otterington at about 4.20. We had to go slowly on account of the fog. It was not a dense standing fog, but seemed to pass in waves. Before I started I could see the up advance-signal from the cabin at Otterington. As we got further along the road the fog got thicker, and I do not think we could see the signal lights for above 100 yards. I took the ganger and three platelayers with me. I went straight to the signal-cabin to see that all was made secure, and learnt that the guard had gone to Avenue junction to block the line. I then came along the down line side of the wreckage. I found two injured people laid alongside the line. I attended to these two. I then searched the wreckage all along that side, and saw the guard's body and two others dead under the Pullman car. I went round the other side and saw two injured and one dead person by the side of the hedge. I went and got the platelayers, who had been at work at the luggage, and we removed these persons into the field, and we got cushions and beds from the sleeping car. Guard Bean was assisting me. I saw no fire break out until after half-past 5. I made inquiry as to whether there was any water to be got, and was told by some of the farm people who had arrived that there was a pond near. I knew there was a water main passing near, but did not know exactly where it was. Holmes did not tell me that he had not been in bed on Monday.

7. *Thomas Pick* states: I have been 19 years in the service of the company, and am district inspector in the traffic department. The line between York and Northallerton is part of my district, and I am directly in charge of the signalmen in that district. On the 1st November a telegram to the following effect was received at my office at York at 2.47 p.m. from Mr. Kirby. "Can you send relief to Manor House cabin to-night, Holmes' child dead." I was away on duty on the Scarboro' line, and my assistant E. L. Harland sent an answer at 4.35, as follows:—"Cannot relieve Holmes to-night." On my return to the office at 5.45 Harland had gone, but I found the copies of the messages on my desk. I looked over my list of relief signalmen, and saw that I had not a man to send, unless I had sent a man who had been on duty. I therefore did nothing further, except that I came back to the station at about 7.30, expecting that if Holmes wanted to say anything further I would have been communicated with. Supposing I had received a telegram that the man was unfit for duty, I would have done the best I could under the circumstances. There were men who had only recently come off duty, and who might have been sent if it had been absolutely necessary, but they would not have had a proper amount of rest. There was exceptional pressure on the signalling department at that time, for, owing to a bridge having been washed down near Temple Hirst, there were five men specially employed there. There are 356 signalmen in the York, and in addition 17 relief signalmen. In ordinary circumstances I have found that to be a sufficient proportion of relief men. I find there are 283 signalmen altogether in the York district, and in addition 17 relief signalmen. They were employed on that day as shown on the return I hand in (*see* Appendix). I have never before had any difficulty in affording relief to a signalman. Even although I did not know that Holmes had stated he was unfit for duty, I would have relieved him if I had had a man available, on hearing that he had lost his child. I have done so before on similar occasions. In an extreme case a stationmaster would be justified on switching out a cabin, but I would only do it myself in an exceptional case. I do not think it is at all a good thing to switch a cabin out without notice if it can be avoided. I should mention that only last week a signalman told me he had on his own responsibility once switched out his cabin, as he did not feel well enough to carry on the work, and that after the matter had been inquired into he was told that he had done quite right. He did not tell me how long ago this was. I would not find fault with a man for doing this, if he could show me that he had good reason for it.

8. *George Bean* states: I have been about 14 years in the service, and a guard for about five years. On the 1st November I came on duty at 6 p.m., worked the 6.25 p.m. train to Doncaster, the 8.15 Doncaster to York, and then the 10.25 York to Newcastle, arriving at 12.33. I was told there that I had to take charge of the second portion of the 10.30 express Edinburgh train, the first portion of which is due to leave Newcastle at 1.42 a.m. The second portion arrived at Newcastle at about 2.30 a.m., and started at 2.38 a.m. It was made up a follows:—Engine and tender, Great Northern brake-van, East Coast third-class, Pullman sleeping car, East Coast composite, East Coast brake-van, East Coast sleeping carriage, East Coast composite, two East Coast third-class, Great Northern saloon, Great North of Scotland horse-box, Caledonian brake-van, and East Coast brake-van on which I was riding. The conductor was in the front van from Newcastle to Darlington, and then he changed into the middle van. The whole train was fitted with the Westinghouse brake, except the Great Northern brake-van, and the Great Northern saloon, which had only the through pipes. The brake was in good order. We stopped at Darlington only. I had no warning whatever of the collision. It was a misty morning, and I cannot speak as to the position of the signals at Otterington or Manor House. I was just getting up off my seat when some barrels of fish were thrown violently across the van, and jammed me against the door. I was not injured, but a little shaken. There was no application of the brake before the collision. I got out as soon as I could, but was a little dazed. The three rear vehicles of the train were still on the rails, but all the others were off. I got back into the van again for my lamp which had been put out. I relit it, and took my signals, and ran forward as fast as I could to the signal cabin. Just as I got there I met the signalman coming down the steps. I said, "Oh, dear! what have you done?" The only answer I could get from him was, "I have a child dead in the house. "Had I been relieved as I asked to be this would "not have occurred." I asked him if he had blocked both lines, and he said he had already done so. After seeing that the home-signal was at danger I went forward to Avenue junction cabin. I saw that the down mail had been stopped there. I asked the guard of the down mail if he would protect his train, and then went and told the signalman what had occurred, and asked him to telephone to Thirsk for medical aid and assistance. I stayed in the cabin till he had done so. I then came back to my own train with the Leeds guard out of the down mail. The accident happened at about 4.3 a.m., and I did not get back to my train till about a quarter or ten minutes to 5. The fire commenced when I was about 40 or 50 yards away from the wreckage. It commenced at the engine. The engine and tender were both upset on their left sides. The two leading vehicles were smashed to atoms; the Pullman car was some distance south of the engine, the body being separated from the bogie frames, and resting partly upon the wreckage of the van and leading carriages. None of the carriages behind the Pullman were upset, but some of them were telescoped, the middle van being completely telescoped into the composite next to the Pullman. It was rather a heavy train. Most or all of the people who were killed were either in the front third-class carriage or in the composite next to the Pullman. I think the fire began in the East Coast van. The relief train arrived, I believe, at about 5.40 a.m. I did not go away with it. I helped the passengers into it. There was no water at hand to put out the fire when it first commenced, and, I think, about six carriages were burnt. The rear carriages were drawn back clear. I left duty on Monday at 9 p.m., and remained off till 6 p.m. the following day.

Richard Wimpress states: I am conductor in the East Coast service, and have been over 26 years in railway service. I was conductor of the 2nd portion of the 10.30 p.m. train from Edinburgh on the 1st November. It started at 11.2 p.m., which was 32 minutes late. We left Newcastle at 2.38 a.m., which was 56 minutes late. We lost six minutes before Berwick by signals, were 11 minutes over time at Berwick, and lost eight minutes by signals between Berwick and Newcastle, where it was very foggy at places. The fog was bad at Durham. After leaving Northallerton it was hazy, but there was not a fog. From Newcastle to Darlington I was in the front van, but at Darlington I changed into the Aberdeen van to get the luggage ready for York. I did not see the signals at Otterington or Manor House as it was frosty and steam was beating down. I had no warning of the collision. It threw me about, and I was bruised by luggage falling on me. I think I was unable to get out for about 10 minutes or so. The gas was put out by the collision. As soon as I got out I went to the signal-cabin. I went into the cabin, and I ascertained that the signalman had blocked the line in both directions. He was very much distressed and I did not question him at all. I got some oil to trim my lamp, and then went to Avenue junction to ask the signalman to send for assistance. After doing this I came back to the scene of the collision, getting

there about three-quarters of an hour after the collision, or a little more. When I got back the fire had commenced. It commenced on the top of the fire-box of the engine, and I think it was the wreckage of the two front vehicles which was lying there and caught fire. By that time several of the bodies had been taken out. There was one child who could not be got at, who was lying below the wreckage which was on fire. I gave all the assistance I could. I think it was about 8 a.m. when the special, which came to take the passengers away, started back to Thirsk. When I got back from Avenue junction, the Aberdeen van in which I had been riding, was not yet burning, nor was the East Coast composite. It was some time afterwards before the fire reached them. I came on duty at 9.45 p.m., having left duty at 9.15 a.m., and was due in London at 8 a.m.

10. *Andrew Baillie* states: I have been in railway service for 23 years, but am a servant of the Pullman Company. I was conductor in charge of the sleeping car "India" on the second portion of the up Scotch express on the night of the 1st–2nd November. I had no warning of the collision. I was in the smoking compartment, but not asleep. I was slightly bruised, but not hurt to speak of. We were running at full speed, 50 or 60 miles an hour, and there was a sudden crash. The Pullman car dashed right through the two carriages in front of it, the engine and tender turning off to the left. The roof of the carriage in front of the Pullman came right into the front of the Pullman, breaking down the partitions as far as the second berths. All the lower berths of the car were full. I called out to the passengers to find out if all were there, and after lighting the lamps I counted them and found all were safe. I then got through a window, went to the signal-cabin, and found that the line had been blocked, and then went back to assist in the best way I could, getting mattresses and pillows from the car, and spirit flasks from the passengers. I then walked round the train, and when coming back from the rear I found the driver of our train lying amongst the coals. He was perfectly composed, and said that his leg was broken. I got a mattress and put him on it, and covered him with blankets, and then he told us to go and look after the passengers. I did not see the fireman, but the driver stated that he had gone back to the rear. The fire broke out in the wreckage on the engine, the wreckage, I think, of the first two vehicles, as I was trying to get out the body of a lady who was, I think, dead. This was the only body I saw under the wreckage, but we heard the cries of a child who was below the woman. I think all the other bodies had been got out by that time. The fire did not commence until nearly an hour after the collision. The Pullman car was off its bogie frames, and the body was down on the rails. I saw a child got out. When I heard its cries the wreckage above and below it was not on fire.

11. *Joseph Barnes* states; I have been 33 years in the service and a driver for 23 years. On the 2nd November I came on duty at Middlesboro' at 1 a.m., to work for 10 hours. I had left duty at 11 a.m. on the 1st November. My engine No. 148 is a six-wheel coupled goods engine, with six-wheeled tender. There are hand-brakes on both engine and tender. I was attached to the 2.0 a.m. up goods train from Middlesboro' to Starbeck. I started at right time. I had to pick up at Welbury. When we left Welbury I had on 35 waggons, all but two or three loaded, and one goods brake-van in rear, in which my guard George Petch was riding. After leaving Welbury we stopped first at Northallerton lower yard. We stopped there for about ten minutes, and then proceeded. We had clear signals at Northallerton station, Hutton Wood, and Otterington. The Manor House up distant-signal was standing at danger. I applied my brakes, and came slowly forward to the home-signal which was also at danger. I came to a stand at this signal. As near as I can judge I stood there for four or five minutes. The signal was then lowered and I started. I think I was running at about six miles an hour, and that about half my train had passed the signal-cabin at Manor House when I felt the collision. The shock was not severe upon the engine, but I felt that something had come into us. The shock shoved us ahead. I ran forward and stopped at the advance-signal. I then got off, and went back to the back end of the train, and saw that the Scotch express had run into my train. My brake-van, and I believe one waggon were broken to pieces, but only one other waggon, and one pair of wheels of a third had left the rails. Some of the couplings had been broken. My guard was killed, but neither I nor my fireman was hurt. The engine and the tender of the express were upset on the left-hand side of the line. The Pullman car was south of the engine, and I think that the body was off the bogie frames. I went to the signalman and asked if he had blocked the road, and he said that he had. He was on the landing outside his cabin. I then went back to the engine. I reckon that my speed on passing Manor House distant-signal was about 20 miles an hour, which speed was gradually reduced. I saw the signal when about 30 yards from it, and I at once shut off steam and applied the brakes. I did not sound my whistle when standing at the home-signal. I did not think we were standing an unusually long time. I did not see the first portion of the Scotch express pass Northallerton, and did not know how it was running. I did not know whether my guard left his van while we were standing. I did not go to the signal-cabin, nor did my mate. We were close to it. It was, I daresay, about five minutes after I came to a stand at the advance-signal that I went back to the point of collision. None of the carriages were then on fire, and the fire had not commenced when I went back to the engine. I came back a second time, and the fire had not commenced then. I went to help the passengers, and remained there for 2 hours 40 minutes. As near as I can judge the fire did not commence until three-quarters of an hour or one hour after the accident. It started from the wreckage on the engine, and gradually extended northwards. There were, I think, two people under the wreckage who could not be got out before the fire reached them, but I believe they were already dead. I left just after a relief train arrived, and went on to Thirsk. It was a foggy night. It was not a thick fog, but I have seen fog-men out on a night better than that. I could see the signal-cabin at Manor House from where I was standing at the home-signal, but I did not see the signal-man. When I was passing the cabin I saw the signalman. He called out and asked me if the distant-signal was on. I said, "Yes," meaning when I passed it. I had one white head-light on my engine; it was burning properly. I stopped close up to the home-signal at Manor House. I could see the cabin plainly from where I stood. I did not think it was so foggy that I ought to have sent my fireman to the signal-box, according to Rule 275A. I did not look at my watch while I was standing, but I did not think I was detained an unusual time. It is not customary for us to whistle unless we are stopped for an unusual time, which was not the case upon this occasion. I do not think that signal-lights, or tail-lights, were at the time visible for over 30 or 40 yards. At the time the tail of my train was struck the advance starting-signal at Manor House was not visible to me. I saw it just after we were struck.

12. *John William Fenna* states: I have been over five years in the service, and about a year a fireman. On the 2nd November I was firing for driver Barnes. I have heard my driver's evidence, and it is correct so far as I know, except that I estimate our speed at the Manor House distant-signal at about 30 miles an hour. When my driver came back to the engine, after going to the scene of the collision, he

sent me forward to Avenue junction to see if the line was properly blocked. The conductor and the guard of the express had gone just before me. I found a down passenger train had been brought to a stand, and that the line was blocked. I went back to my engine and looked to her, and then went back to the wreck and gave assistance. When I got back from Avenue junction the fire was just starting. I heard the guard say that that was just about an hour after the collision.

13. *William Weighall* states: I am a ganger of permanent-way, in charge of the length at which the accident happened. I live near the station at Otterington. I was called at about 4.45 on the morning of the 2nd November, and went at once to the scene of the accident. I found that 105 yards of the up line was broken up. From what I could see I should think that the tail of the goods train was struck 50 yards or more outside the home-signal. When I got there the train was on fire.

14. *John Bradford Harper* states: I am station-master at York. I was called at 4.50 a.m. on the 2nd instant and informed of the accident. I went with a special which left York at about 5.30, taking doctors and ambulance men with appliances. We got to Thirsk north junction at 6.18, and had to wait there for the pilot-man. We got to Manor House about 6.50. Six of the bodies had then been removed from the wreckage. The wreckage of the leading van, a third-class carriage, and a goods waggon were immediately behind the body of the Pullman, which was off its wheels and a little south of the engine. Partly on the engine and on the wreckage was a composite carriage, and underneath the north end of the composite and standing foul of the down main line was a sleeping carriage. Sitting on the top of the two was the Aberdeen van, the leading end of this van being telescoped into the top of the composite. The next vehicle was a third-class carriage with the leading end driven up underneath the van, and the next was a composite carriage which had telescoped with this third-class carriage. The next was a third-class carriage off the rails, and then a saloon off the rails, and the next three were on the rails. When I arrived the wreckage of two leading vehicles and the waggon were well on fire, and so was the composite which was on them and the engine, and the Aberdeen van was just taking fire at the leading end. I set my men to search the carriages which were not on fire. They were gradually driven back by the flames. We found no passengers in these carriages, but some dogs in the Aberdeen van could not be got out. We found two dead bodies under the leading end of the Pullman. The child had been got out before I got there. The train I brought went away with the passengers at about 7.30. They had to be collected from the cottages and farms. They came to Thirsk, where some had already arrived. They all went south in the train at about 8 a.m., and by a special to London from York at 9.15. The breakdown trains from Darlington and York arrived soon after I did.

15. *Luke Welch* states: I am station-master at Thirsk. I was called to the scene of the accident at about 4.20, and got there at 5.40. I had been occupied in making arrangements for taking the passengers away. I made up a special of a carriage and van, and took down as many people as I could to assist. When I got there I found the train was on fire, but only about two carriages were alight at that time. I saw a child got out from the wreckage, which was partially on fire. She was alive, but was rather badly burnt on the feet and legs. She was apparently insensible, and was taken back to Thirsk in my train. I am told she died on the way. The passengers, or as many of them who would come, were brought back to Thirsk in the down mail, which had been stopped at Avenue junction. The down mail went north by another route. I do not know how the up passengers were taken forward.

16. *Roland Ewart*, states: I have been nearly 18 years in the service, and a driver for eight years. I am well acquainted with all the line between Berwick and York. On the 1st November I came on duty at Gateshead at 10 p.m. to work till 8 a.m. I had left duty at 9 o'clock in the morning of that day. My engine, No. 178, is a tender-engine with six wheels, the driving and trailing wheels being coupled. The tender has six wheels. The engine is fitted with a Westinghouse brake, working blocks on the four coupled engine wheels, and on the six tender wheels. I left Newcastle at 2.38 a.m. with the second portion of the up Scotch express. There were 13 vehicles in the train, equal to 15. The Westinghouse brake was working throughout the train, 10 of the vehicles being fitted with the Westinghouse brake, and the others with through pipes. The brake was properly tested before starting and it was in good order. I used it to slack the train at Durham and Thinford junction, where I nearly brought the train to a stand, and at Darlington where we stopped, and it acted properly. We left Darlington at 3.39 a.m. and the signals being right all the way, after that I did not use the brake again. I was running with a pressure of 65 lbs. as shown on the brake-gauge. The weather was very thick from Newcastle to Durham, but the signals as far as Darlington were clearly visible for 50 yards. After Darlington it was pretty clear till Dalton junction, and the signals were visible for about 150 yards. It then got rather thick to about Danby Whisk. It then cleared up again a little, and it was pretty clear at Northallerton where I had no difficulty in seeing the signals 200 or 300 yards off. After passing Northallerton it came on very thick, thicker than we had had it all through the journey. The signals at Hutton Wood were not visible to me until I was within 20 or 30 yards of them. They were off. The Otterington signals were all off, when I sighted at about 20 or 30 yards distance. The Manor House up distant-signal was off. I sighted that when about 20 or 30 yards from it. I passed this signal at my usual speed which would be 55 miles an hour, or perhaps a little more. I was doing my best to keep time, which I was nearly doing. I expected to have lost about 1½ minutes or two minutes when passing Thirsk. I was looking up and a-head for the Manor House up home-signal, but I did not see it. My proper side is the right side. The distant-signal is on the left side, and I had gone to that side to look for it, and I then crossed to my own side to look for the home-signal. While I was looking for it I caught sight of three red lights on the line ahead of me, not above 20 or 30 yards off. I was looking over the side of the engine, outside the cab, and I had no time even to shut off steam or to get to my brake-handle before I was knocked over against the tender. I do not remember how I was thrown from my engine, but when I came to myself I was lying in the field outside the fence of the line, three or four yards south of the home-signal and five or six yards inside the field. I do not know where the engine was, or in what position it was lying, but I could hear the steam escaping. I saw the fire commence some time after the accident, perhaps half-an-hour or more. It commenced quite gradually like the flicker of a candle, and I could have put it out quite easily if I had been able to get to it. The first man who came to my assistance was the Pullman car conductor. He came before the fire commenced. He gave me a little brandy, and I told him to go and look after the passengers. Some time after that he brought me some cushions and moved me further back into the field. The guard Bean came to me some time afterwards, and one gentleman came with him. At that time the fire was burning strongly. Several people had come to look at me, and I had told them they could do me no good, and they had better go and

assist others. The fire commenced nearly opposite to me, but on the other side of the line. I had the idea that the fire had commenced from the stove in the goods-van. I cannot estimate how far the engine ran after striking the goods train before it turned over. The fog was varying and lifting at times, so that I would not have expected to find fog men out. When I came to myself, after being thrown from the engine, I could see the arm of the Manor House up home-signal. It was still off, and I saw it go to danger. I did not know what train it was I had struck until I was told. I thought at first it was the first portion of the express. My fireman was looking out for signals at Otterington, but after passing Otterington he was attending to the fire and to the injector. Manor House is switched out on Sundays, but I have never known it switched out at other times. I have never known a box switched out except in regular course. When a box is switched out the lights are extinguished, except at junctions. The fog that night was so thick that I would have considered it to be my duty to send to the signal-box if I had been stopped at the home-signal, unless I could see the signalman. I could see the windows of the signal-box from where I was lying.

17. *Edward Head* states: I have been about seven years in the service and 2½ years a fireman. I am a spare fireman and work where required. I have not very often run from Newcastle to York, but only when the Scotch express is divided, or with specials. On the 1st November I was fireman with driver Roland Ewart, and came on duty with him at Gateshead. I had left duty at 8·30 a.m. on the 1st November. It was very foggy all the way from Newcastle to Manor House. Occasionally it was awfully thick, so thick that the signals were not visible until we were close underneath them. I cannot say where it was thickest, because I did not often look up for the signals, being otherwise engaged. I was firing nearly all the way and keeping up steam. I did not see the signals at Northallerton as I was firing. I saw none of them. It is part of my duty to look after signals, but when I am otherwise engaged I cannot do so. I have a rule book. I do not know Rule 284. I have read the rules and know that "the "engine driver must, as far as practicable, have his "fireman disengaged when approaching or passing a "station or junction, so that he also may keep a "good look out for signals." I did not look out for the signal at Hutton Wood or Otterington, nor did I see them. I did not see either the up home or the up distant-signal at Manor House. I did not see the tail lights of the train which we struck. I did not hear or see anything unusual until we struck the other train. I was on my proper side of the footplate, the left side, with one foot on the tender and the other on the engine. I had been firing and was just lifting my back after using the shovel. I do not know how I was thrown off the engine, but when I gathered myself up, I was lying on my back inside the railway fence near the Pullman car, south of the engine. I was able to get up. I went at once of my own accord back towards Otterington. I went to protect the train. I was not able to get as far as Otterington. I went to a farm-house near the line. I do not know quite how far I had gone. Perhaps an hour afterwards I was taken back by Mr. Kirby to the signal-cabin at Otterington, and then to Mr. Kirby's house. I was moved the same day to the hospital at Northallerton, where I remained for about 15 days. I am still feeling a pain in my back. I think the speed of the train at the time the accident happened was between 50 and 60 miles an hour. As I was walking back towards Otterington I did not look at the up signals. I looked at the down signals. They were at danger. I could see them plainly. I could not say the exact distance I walked after starting for Otterington before I saw the down signal, but perhaps 600 yards. The signal was then about 12 yards from me. I did not look to see if I could see Manor House cabin from where I was lying. From where I was lying the carriages and Pullman car hid the Manor House up home-signal from me.

18. *Joseph Head* states: I am in the locomotive department of the North-Eastern Railway Company, and have charge of the brakes and gas apparatus upon the carriage stock. I hand in a statement (*see* Appendix) showing the class of gas fittings, the dimensions of the reservoirs, the pressure in each when the train left Newcastle, and the probable pressure at the time of the accident. Pope's reservoirs, when fully charged, are charged at a pressure of 105 lbs., and Pintsch's at 90 lbs. The vehicles would have been fully charged at York, and when fully charged the gas should last for 24 hours. Each vehicle is self-contained. We have tried some experiments to see how long it takes for the gas to escape, and I hand in a return showing the result. The filling holes are about ⅜" in diameter. I made an examination of the stock after the accident, and made a report to Mr. Worsdell, which is embodied in Mr. Worsdell's report of November 30th, 1892, now produced. (*See* Appendix.)

19. *John Welburn* states: I am superintendent of the line on the North-Eastern Railway. I hand in the following statement, showing the arrangements which were made at Otterington, Thirsk, and York for the relief of the passengers in the second portion of the Scotch express after the accident. This collision occurred as nearly as can be ascertained at 4.2 a.m., and on signalman Eden, who was on duty at the Otterington station cabin, having been informed of it by telegraph from Manor House, he immediately called up Mr. Kirby, the station-master, and afterwards four platelayers from the cottages near the station. Mr. Kirby arrived at the cabin at about 4.20, accompanied by his son, who was at once sent to the village to call up the porter and some of the Company's workmen. Mr. Kirby sent messages to Thirsk, Northallerton, Darlington, and York, and left the station at 4.30 a.m., accompanied by the men he had got together, on an engine of a goods train which had been stopped at Otterington. On receipt of a telegram at Manor House, from Northallerton, to say that a special with medical men and other assistance had left at 6.10 a.m., Mr. Kirby went back to Otterington to clear the line for this special. On receipt of a telegram at High junction cabin, Northallerton, the night porter was sent by the signalman to call the station-master and the assistant station-master. Mr. Beecroft (the station-master) was at the station at 4.50, and immediately sent for Mr. Baines, the permanent-way superintendent. He also got the postman, who was at the station, to call the doctors, whilst the station staff called the platelayers and others at the Low junction, and at Normandy. An engine was got round from the shed on the Melmerby branch, and with it a coal train standing at Hutton's Wood, was brought back to Northallerton. By this time three medical men had arrived, and the party left Northallerton at 6.7, stopped at Otterington, and arrived at Manor House at about 6.25. Information of the accident was received at Thirsk at 4.10 from the signalman at Avenue junction. The night assistant station-master, Thackray, who was on duty, had the station-master, the assistant station-master, and other servants of the Company, residing in cottages close by, at once called up, and a shunter was sent to Thirsk Town to summon Drs. Hartley and Ferguson, who drove direct to Manor House, reaching there at about 5.45. The permanent-way inspector, who lives at Green Lane, was called up; an engine, van, and carriage got ready, and after a message had been sent to Darlington, Northallerton, and York, it left for Manor House at 5.0, but as the first part of the mail was standing at Avenue junction, they took it forward, leaving at 5.29, and arriving at Manor House at about 5.35. At 6.30 the

first train (the mail taken from Avenue junction to Manor House), left for Thirsk with a few passengers who had not gone to the farm and cottages close by, and four dead and four serious cases of injury, including driver Ewart and Mr. Buchanan. Two of the injured died on the way to Thirsk, and the train was run forward to Thirsk Town station for the purpose of getting Mr. Buchanan and driver Ewart to the cottage hospital at Thirsk as early as possible. The other passengers remained at Thirsk for the second special which left at 8.1 a.m. A train taken from York returned as the second special with passengers leaving the Manor House at 7.23, and on arrival at Thirsk the doctors, who accompanied the train, attended to the passengers, refreshments were provided, and the train left at 8.1, arriving at York at 8.30, and after further attention given by Doctor Gosling to some of the passengers, and refreshments, a train left special for London at 9.14 a.m., reaching King's Cross at 1.30. It was necessary to change the passengers at York into a train made up of carriages fitted with the automatic vacuum-brake used on the Great Northern Railway. Information of the accident was received at York at 4.39, and messages were at once sent to the officials on the list to be called in case of accident, and the locomotive foreman was advised by telephone to get the brake-down gang ready. Mr. Harper and Mr. Steel were called at 4.50, reached the station at 5.12, and left by a special train, which had been got ready by the night station-masters at 5.35, having been delayed a few minutes waiting for Drs. Anderson and Gosling. Twenty-eight men, a number of whom belonged to the ambulance corps, with stretchers, &c., were taken by the train. This train stopped at Thirsk, where arrangements were made to work the traffic over the Melmerby branch. The train reached Thirsk at 6.10 (where more assistance was taken up), was delayed at the North junction waiting for the first train coming from Manor House, and reached the latter place about 6.50. The wrecked train was then on fire. The break-down train left York at 5.52, reaching Thirsk at 6.54, and Manor House at 7.5. The Darlington tool-vans arrived at 7 o'clock. A telegram was received at York from Thirsk at 6.8 a.m. that the train was on fire, and the Company's fire engine and brigade were despatched special, leaving at 7.20 and reaching Manor House at 8.18. Water was obtained from a very small stream about 300 yards away, which was dammed up, and a sufficient supply of water was got to put out the fire. The down line was open at 1.3 p.m. The single line working between Avenue junction and Otterington was commenced at 1.3 p.m., and both lines were open and traffic resumed at 9.3 p.m. The whole of the wreckage was cleared away on Sunday the 6th November. The position of the engine and vehicles of the passenger train are given below :—

North-Eastern engine, No. 178.—The engine and tender lay wheels together, the front end of the engine and the back end of the tender resting on the siding rails.

Great Northern van, No. 1057 - -	These vehicles were broken and burnt.
East Coast third, No. 60	

Pulman car, "India."—The front end completely broken away, and the car, without wheels, was found resting with its front end on the rails of the siding, and the rear on the rails of the up main line.

East Coast composite, No. 26 . - -	These vehicles were burnt.
East Coast van, No. 51	
East Coast sleeping, No. 106 - -	
East Coast third, No. 163 - -	
East Coast composite, No. 129.—Partially burnt.	
East Coast third, No. 41 - - -	These vehicles returned to Northallerton.
Great Northern saloon, No. 8 - -	
Great North of Scotland horse-box, No. 13 -	
Caledonian van, No. 59	
East Coast, No. 53 -	

Ten passengers were killed, the bodies of eight were recovered, and two were burnt, and I have, up to the present, received advice of 36 passengers injured. The passengers who were killed were travelling in the third-class carriage immediately in front of the Pullman car. There is no doubt, I think, that the train was set on fire from the engine and tender, and not from the gas used for lighting the carriages.

Conclusion.

The causes of this terrible collision, which had for its result the death of 10 persons, the injury of 43 other persons, and the complete destruction of seven passenger vehicles and nine goods vehicles, are very clearly indicated by the foregoing evidence.

On the night of the 1st November the up Scotch express, due to leave Edinburgh at 10.30 p.m., was run in two portions, the second portion, on account of the late arrival of one of the north trains, not starting until 11.2 p.m., 32 minutes after the first portion.

Owing to foggy weather both portions of the train lost time on the journey, the second portion rather less than the other, and the first portion, which had left Newcastle at 2.15 a.m. on the 2nd November, and Darlington at 3.14 a.m., passed Northallerton north junction at 3.33 a.m., Hutton Wood at 3.35 a.m., Otterington at 3.37 a.m., Manor House at 3.38 a.m., and Avenue junction at 3.40 a.m., all the block-signals for this train having been given and accepted in accordance with the regulations.

At 3.36 a.m. the signalman at Northallerton north junction, after receiving the "Line clear" signal from Hutton Wood for the first portion of the express, and having been informed by telegraph that there was more than the requisite margin of 20 minutes before the second portion of the express was due, allowed the goods train from Middlesborough to Starbeck, which had been waiting at Northallerton low junction, to come on to the up main line, and to proceed upon its journey; and this train passed Northallerton north junction at 3.38 a.m., Hutton Wood at 3.43 a.m., and Otterington at 3.47 a.m. The signalman at Otterington, who, according to his train record-book, had received from Manor House the "Line clear" for the first

portion of the express at 3.38 a.m., gave the "Be ready" signal to Manor House for the goods train at 3.43 a.m., as soon as he received the "On line" signal for this train from Hutton Wood, and, the signal being accepted, the train was allowed to pass; was given "On line" to Manor House at 3.47 a.m., and, having passed the up distant-signal at danger, came to a stand at Manor House up home-signal, 57 yards north of the signal-cabin, at about 3.55 a.m., according to the driver's estimate, but probably some minutes sooner, judging from the time that the train should have taken to run the distance of 1 mile 30 chains from Otterington.

There is no entry in the Manor House train record-book of any of the block-signals for the goods train, the last entry in the book being the "On line" given to Avenue junction for the first portion of the express at 3.38 a.m., the "Line clear" from Avenue junction for this train, which was sent from Avenue junction at 3.40 a.m. not being entered. The signalman, however, admits that he received, and accepted, the "Be ready" for the goods train, that he did not give any signal forward to Avenue junction for it, and that consequently he did not lower his up-signals, which had been put to danger after the passing of the first portion of the express.

At 3.58 a.m. the signalman at Otterington received from Hutton Wood and accepted at once, the "Be ready" signal for the second portion of the express, which had left Newcastle at 2.38 a.m., and Darlington at 3.39 a.m., and passed Northallerton north junction at 3.58 a.m., and Hutton Wood at 3.59 a.m., at which time it was given "On line" to Otterington.

The Otterington signalman gave the "Be ready" for this train to Manor House at 4 a.m., and immediately got back the "Attention" signal, then the "Line clear" signal for the goods train (as he thought, and was entitled to think), and then the acceptance of the "Be ready" signal for the second portion of the express, whereupon he took off his signals for this train, which passed at 4.1 a.m., running at full speed, probably at nearly 60 miles an hour. Almost immediately after he gave the "On line" signal he was called on the speaking telegraph instrument by the signalman at Manor House, who asked him whether it was the E. P. (express passenger train), which he had given "On line," and he answered "Yes," and then, upon going to the window of his cabin to look out, he heard the crash of the collision.

The signalman at Manor House, James Holmes, states that he remembers getting the "Be ready" signal for the goods train, and accepting it, but he did not record this or any other signal for the train, does not remember receiving the "On line" signal for it, which according to the Otterington signalman was properly acknowledged when sent at 3.47 a.m., and being, as he says, overpowered by sleep, was not aware of the arrival of the train at his up home-signal.

Upon receiving the "Attention" signal and the "Be ready" for the second portion of the express at 4 a.m. he first gave back the "Attention" signal, and then the "Line clear" signal, thinking that he had omitted to give this signal back for the first portion of the express, and he then accepted the "Be ready" signal and lowered his up signals; upon which the goods train started, had run forward for its whole length of 232 yards, less 35 yards, and had attained a speed of about six miles an hour, when the express ran into it at full speed, with steam on and the brakes not applied, and he disastrous consequences recorded at the commencement of this report ensued.

Such being a brief summary of the events leading up to this collision, it is perfectly clear that it was principally due to non-compliance with the block telegraph rules and regulations by signalman James Holmes, who had at the time been on duty for eight hours, and who was manifestly unfit for duty from causes which will be dealt with hereafter.

The actual breaches of the rules by signalman Holmes were as follows, viz.: He omitted to send forward to Avenue junction the "Be ready" signal for the goods train after he had received it "On line" from Otterington; he omitted to book the "Line clear" signal received from Avenue junction for the first portion of the express, and the "On line" signal received from Otterington for the goods train; and he sent the "Line clear" signal to Otterington, and accepted the "Be ready" for the second portion of the express, when the goods train was standing with the head-light of the engine within his view just outside his up home-signal.

The signalman, however, was not the only person who acted in contravention of the rules, for it was, in my opinion, clearly the duty of the driver of the goods train, Joseph Barnes, when brought to a stand at the up home-signal at Manor House, to sound his whistle before remaining there for a length of time, which he estimates at from four to five minutes, but which was probably nearer seven minutes, considering

that a period of 13 minutes from the time that he passed Otterington had elapsed before he started from Manor House up home-signal, and that it is most improbable that he had taken more than six minutes to run the distance of under 1½ miles from Otterington, making every allowance for the fact that after passing the distant-signal, which he did at a speed of between 20 and 30 miles an hour, he was pulling up his train so as to stop at the home-signal. If this driver had obeyed Rule 275A (quoted above), he would certainly have sounded his engine whistle, and thereby probably have roused the signalman, while, considering the state of the weather, it is a question whether the last paragraph of this rule should not have been acted upon, and the fireman sent at once to the signal-cabin, and, therefore, I think that part of the responsibility for this collision must be borne by driver Barnes, who had at the time been upon duty for three hours. He states in excuse, that it is not customary for drivers to whistle, unless they are detained for an unusual time, which he did not think was the case on this occasion, and, if his statement be correct, it is most necessary that drivers should be warned that this rule, which seems to me to be quite clear and explicit, must be more strictly attended to.

The only other servants of the Company on duty at the time, whose conduct could possibly be held to have contributed to the accident, were the driver of the second portion of the express, the signalman at Otterington, and the signalman at Northallerton north junction. If some of the evidence as to the state of the weather be correct, the driver of the express, Roland Ewart, ought certainly to have seen the tail lights of the goods train at a sufficient distance to have enabled him, by the use of his continuous-brake, to reduce very much the speed of his train, or even possibly to stop it short of the point of collision. The evidence, however, as to the distance at which signal-lights were visible at the time is so conflicting, and the fact that the fog, which undoubtedly prevailed to some extent, was very variable is so clearly established, that I hesitate to discredit the driver's statement that he was keeping a good look out, and yet did not see these tail-lights until within 20 or 30 yards of them; although he has evidently somewhat under-estimated the distance at which these lights were visible, for if he could, after the accident, see the signal-cabin when lying 50 yards away from it, as he states he did, he should certainly have seen the tail-lights for at least this distance. Even if he did see them so far off and no farther, he had hardly time to do much to stop his train after realising his danger, for travelling at a speed of nearly 60 miles an hour, he would have had under two seconds in which to take action. Driver Ewart, who, with his fireman, had a most wonderful escape, gave his evidence in a very straightforward manner, and seems by all accounts to have behaved very pluckily after the accident.

The signalman at Otterington, Henry Eden, whose train record book, I may observe, was very badly kept, appears to have carried out quite correctly the duties of signalling the two portions of the express and the goods train, but if he had been quite alive to the importance of his office, and especially if he had kept in mind the condition of signalman Holmes when he went on duty, he ought to have taken steps to ascertain from Manor House why he did not receive "Line clear" for the goods train before 13 minutes had elapsed, seeing that the train had only 1 mile 30 chains to run, and he had every reason to believe that, unless there was some break down, it would have a clear run to Avenue junction, where it would run off the up main line on to the independent or goods line. He is perfectly candid in his statement, and allows that if it had occurred to him he certainly would have inquired, adding that, after sending the "On line" signal for the goods train, he sat down by the fire to read, a perfectly legitimate act, but one which would have been better deferred until the block signalling in connection with the goods train had been completed. This signalman, who had been on duty for 8¾ hours, did not, however, infringe any of the rules of the company, and can hardly be blamed for what occurred.

The action of the signalman at Northallerton north junction in allowing the goods train to come off the branch on to the main line, between the two portions of the Scotch express, has been much criticised, but an examination of the running record of the goods train and the second portion of the express shows that he was quite justified in doing so, as there was more than the regulated margin of 20 minutes between the two trains, the second portion of the express not leaving Darlington until one minute after the goods train had left Northallerton north junction, and, as the distance from this junction to Avenue junction, where there are facing points leading on to a running goods line, is only 6 miles 11 chains, I should consider that the margin allowed is sufficient.

There was ample time for the goods train to have reached Avenue junction, only 1 mile 18 chains south of Manor House, and to have cleared the main line, some minutes before the second portion of the express was due there, and it is difficult to see how traffic could be carried on if through goods trains were not permitted to take advantage of an interval of 20 minutes to run a little over 6 miles.

Moreover, the line being worked on the absolute block system, it matters nothing, so far as risk of collision is concerned, whether a train running in advance of an express train is a goods train or a passenger train, for, if the block regulations are attended to, the worst that can happen is some delay to the express.

As before stated, signalman Holmes was honest enough to admit at once that, after accepting the goods train, he fell asleep, and remembers nothing that happened until he got the attention signal for the express, and, as it seems clear that he was unfit for duty, it is necessary to inquire how it happened that he became so, and why, such being the case, he was allowed to go on duty at all.

As may be gathered from his statement, which is uncontradicted, and from the other evidence, he was not on duty at all on Sunday, October 30th, and employed himself on Monday, October 31st, in working in his garden instead of, as usual, taking some rest before going on night duty. He went on duty at 6 p.m. on Monday, and came off after 12 hours work at 6 a.m. on Tuesday, November 1st, after which he walked to his home, a distance of 7½ miles, arriving at about 7.15 a.m. Here he found that one of his children was ill, and, after getting a certain amount of sleep between 8 a.m. and noon, he was roused by his wife, and told that the child was in a fit, and he then went for a doctor, walking about 1 mile to Otterington station, going on by train to Northallerton, where he was unable to find the doctor, and then walking home, a distance of about 4½ miles, arriving at about 2 p.m. to find that his child was dead. He then walked again to Otterington to telegraph for his mother to come to be with his wife, and on his way went to the station to tell Mr. Kirby, the station-master, what had happened, and to request that a telegram might be sent to York to ask if he might be relieved from duty that night, as he knew that "he was not fit for duty." After this he walked home again, returning to the station at 6 p.m, to find that the answer received from York was that he could not be relieved, whereupon, by permission of Mr. Kirby, he arranged with signalman Creed to remain on duty for two hours longer, while he returned home to get some neighbour to look after his wife, because his mother had not yet arrived, as he had expected she would. He eventually went on duty at 8 p.m., having on his way gone to the signal-cabin at Otterington to ask the signalman there to telegraph to let him know if his mother arrived by a later train, remarking that he would go on duty although he did not feel fit for it, or that he was "about done up," and after getting to his signal-cabin he carried out all his duties quite correctly until after the first portion of the express had passed at 3.38 a.m., when he says that he became quite confused all at once, although he afterwards accepted the goods train at 3.43 a.m., and correctly acknowledged the "On line" signal for this train at 3.47 a.m.

Mr. Kirby confirms Holmes' statement that, at their first interview at about 2 p.m., he said he was "not fit for duty," but he did not mention this to traffic inspector Pick in his telegram to York, sent at about 2.47 p.m., which was in the following words, "Can you send relief to Manor House to-night, Holmes' child dead"; the answer, received at about 4.37 p.m., being, "Cannot relieve Holmes to-night." Mr. Kirby states that Holmes, when told this at about 6 p.m., said it was a bad job as he was in "bad fettle for work," adding, "I do not care so much for myself, but I am more "bothered about my wife as she is not fit to be left all night by herself," and then said that, after going home to make some arrangements for the comfort of his wife. he would come to meet the half-past 7 train to see if his mother had come, and would then go on duty.

Now from these statements it appears that Holmes had only a broken rest of four hours after 12 hours duty, that he was on his legs more or less all day, although the actual distance he walked was not over 15 miles, and that he was suffering considerably from mental distress, which facts, if they cannot be accepted as a sufficient excuse for his subsequent failure to perform his duty after undertaking to do it, go a very long way towards accounting for it.

I have no hesitation in saying that he ought to have been relieved, and I am of opinion that it was a grave error of judgment on the part of the Otterington station-master, Mr. Kirby, not to have stated in his telegram to inspector Pick that signalman Holmes had said that he was "not fit for duty." Probably

he thought that the poor fellow would have been relieved for the reason he did give, viz., the death of his child, as no doubt he would have been if there had been a man available; but when the refusal to relieve him was received, Mr. Kirby might easily have made a fresh appeal on the ground of Holmes being unfit for duty.

It must, however, be remembered that it is comparatively easy to say after an event what should or should not have been done, that when signalman Holmes came to the station at about 6 p.m., he seemed to be more troubled about his wife than about himself, as is stated by signalman Eden as well as by Mr. Kirby, that, although he told Mr. Kirby he was "in bad fettle for work," a somewhat less emphatic expression than saying he was "unfit for work," he voluntarily, and in a very proper spirit, went to make arrangements for going on duty, and made no fresh request to be relieved, and that Mr. Kirby formed the deliberate opinion, wrong though it turned out to be, that signalman Holmes was in a fit state to go on duty, and took care that his anxiety should be lessened by the information that his mother had arrived, and had gone to look after his wife. It is a fact that it was within the discretion of Mr. Kirby to have "switched out" Manor House signal-cabin, that is to have closed it for the night, but it is not advisable to close signal-cabins without notice if it can possibly be avoided, and if he really thought, as he says that he did, that signalman Holmes might safely go on duty, it would not have been right for him to do so.

The refusal to relieve signalman Holmes was sent by a clerk in the absence of inspector Pick, but was practically endorsed by the latter, and under the circumstances I cannot see that he could have done anything else. If it had been reported to him that the signalman had stated that he was unfit for duty, he ought certainly to have taken steps to relieve him, or to switch out the cabin, but as all his relief-men were employed he could only have relieved Holmes by sending a man who had not had his proper allowance of rest after coming off duty, and, however much he may have sympathised with the signalman in his bereavement, as everyone must do, this was not by itself a sufficient reason for incurring the risk of sending on duty a man only half rested, or for switching out the cabin without notice.

Having dealt with the events preceding the collision, it becomes my duty to comment upon the action of the servants of the Company after it occurred, and here it must not be forgotten that in a catastrophe of this magnitude there must inevitably be a great amount of confusion, in which only an exceptional man would act in a perfectly cool and collected manner. Still, when there is a plain rule laid down for the guidance of those concerned it should be obeyed, and therefore I must report that, after finding out what had taken place, it was the duty of driver Barnes (under Rule 281, of which a copy is given above) to detach his engine and to run forward to Avenue junction, instead of sending his fireman or allowing him to go there. Much valuable time would have been gained thereby, and the other servants of the Company who walked to Avenue junction might have been more usefully employed at the scene of the accident.

As it was, with the exception of the Pullman car conductor, Andrew Baillie, who seems to have done everything in his power, there was not one of the men belonging to the train in a position to give assistance to the passengers until nearly an hour had elapsed. There would not have been the slightest necessity for either the conductor or the guard of the train to go to Avenue junction signal-cabin, if the engine of the goods train had run forward, and it is difficult to understand how both of these men, as well as the fireman of the goods engine, could have gone to Avenue junction without meeting. No doubt they acted, on the spur of the moment, in the manner which they thought was best calculated to safeguard the line, and that being their first duty, I do not think that they should be blamed for what was at most a mistake.

There seems also to have been for some length of time an absence of any organisation in the work which was being done at Manor House, for otherwise it is not easy to see why the fire, which did not commence for nearly an hour after the accident, or nearly half an hour after station-master Kirby and four platelayers arrived on the scene, and which driver Ewart describes as being at first a mere flicker which he could have beaten out with a shovel, was not extinguished before it came to a head. It should here be mentioned that the evidence makes it quite clear that the fire was not in any way due to the fact that the train was lighted by compressed gas, that it is nearly certain that it was caused by the wreckage catching light from the engine fire, and that it is hoped and believed that no deaths are to be attributed to the fire, although two bodies were burnt, and one poor little child, who afterwards died, received injuries from burns before she could be extricated from the débris.

It was unfortunate that signalman Holmes did not at once throw his signals to danger when he saw the goods train advancing, instead of losing time by telegraphing to Otterington, for it is just possible that if he had done so the driver of the express might have found the distant-signal at danger when he reached it, which would have enabled him to avert the collision, or at any rate to mitigate its violence. That the signalman is mistaken in saying that he did throw his signals to danger at once is shown by the evidence of driver Ewart, who actually saw the signal-arm of the up home-signal go to danger when he was lying upon the ground after the accident. One feature of this disaster which should be alluded to is in connection with the marshalling of trains, for it brings prominently to notice the safety which is insured to passengers riding in vehicles like the Pullman car, and the danger caused to passengers using smaller and lighter carriages in front of such a vehicle, for the whole of the passengers who were killed on this occasion were in the third-class carriage in front of the Pullman sleeping-car, and from this carriage very few, if any, appear to have escaped without injury. A strong argument is thus furnished for the making up of trains throughout with vehicles of the same pattern and weight, or at any rate for placing such vehicles together in the train.

It now remains to consider what should be done in order to guard against accidents of this class, which is really by far the most important part of the duty of an inspecting officer in these investigations.

It is well known that there are in existence several combinations of electrical and mechanical appliances, the use of one of which, so long as everything was maintained in good order, would make it impossible for such an accident as this to occur, unless a driver were to run deliberately past fixed signals; because, until a train has actually passed over a treadle, or contact-maker, placed in advance of a home-signal, in any position which may be considered suitable, the signalman in the cabin from which this home-signal is worked is prevented from accepting on his block telegraph instrument a following train from the cabin in rear, and is prevented from releasing the starting-signal at the cabin in rear, which signal is electrically controlled by him. It is true that if a train becomes divided between two block-signal cabins, and the driver runs forward without perceiving it, the passing of the engine over the treadle would place the signalman in the position of being able to accept a following train, but there would not be any risk of accident unless the signalman neglected his duty by not looking for the tail-light on the passing train, and unless the guard in charge of the portion of the train left behind in the block section also neglected his duty by omitting to go back to protect his train.

Now, although the mistakes made by signalmen are remarkably few, considering the enormous number of train signalling operations performed daily, it is a fact that these mistakes, when made, frequently lead to very bad accidents, and it is therefore the duty of all railway companies to adopt, upon their important lines at any rate, some safety appliance which will prevent a mistake made by one signalman from causing an accident, if there be such a machine which has been proved to be reliable, and, as some of the inventions which I have mentioned have been in general use for some years upon three lines of railway, and in partial use upon others, both in the United Kingdom and abroad, and have been found satisfactory, I would strongly urge, not only the North-Eastern Railway Company, but other railway companies also, to give this question their immediate consideration.

The fitting up of railways in this manner must, however, be an operation of time, and there is one simple alteration in the block-telegraph regulations of the North-Eastern Railway Company, which would, I am convinced, conduce greatly to the safety of passengers travelling on this line, and which ought to be adopted at once.

Upon some railways, although by no means upon all, unfortunately, a signalman is not allowed even to offer the "Be ready" for a second train until he has received from the cabin in advance the "Out of section," or "Line clear" signal for the preceding train, but this is not the case on the North-Eastern Railway; whereas it is manifest that in this particular case, if such a regulation had been in force, the signalman at Otterington would not, or should not, have offered the "Be ready" for the second portion of the express, but would first have inquired from the signalman at Manor House what had become of the goods train, and the accident would probably not have happened.

Another point to which the attention of the company should be directed is the question of relief signalmen. It was stated, and I have no reason to doubt it, that the proportion of relief signalmen now employed in the York district of the North-Eastern Railway, viz., 17 out of a total number of 300, or just under

6 per cent. has hitherto been found to be quite sufficient, but a case has now arisen in which it was found to be impossible to relieve a man, who, admittedly, ought to have been relieved, and, although the circumstances may have been exceptional, it is clearly the duty of the Company, who alone have the information to enable a judgment to be formed of what is required, to take such steps as may be necessary to prevent such an occurrence in future.

In conclusion, although the length of the hours of work had nothing whatever to do with this accident, and although the causes of signalman Holmes being unfit for duty were matters entirely outside the control of the railway company, I think it right to record my opinion that a tour of duty of ten hours is the outside which ought to be allowed in main line signal-cabins, and that the employment of signalmen for a greater number of hours, consecutively, especially at night, upon lines where they have to deal with considerable traffic of all kinds, including fast express traffic, is an element of danger, even in a signal-cabin as small as that at Manor House, where the working is as simple as it can possibly be in any main line cabin, which is passed by between 80 and 90 trains each way during the 24 hours. It is true that, so far as I can ascertain, there have only been two accidents or collisions in the last 15 years which have been attributed to the action of a signalman waking out of a state of sleep, and none to the fact of a signalman having fallen asleep and remained so; but there is, no doubt, considerable danger when a signalman who has fallen asleep, or who has been stupefied by fatigue or mental distress, as signalman Holmes seems to have been, is suddenly roused, and takes up his work before he is himself again, and therefore every possible precaution should be taken to prevent such a state of things arising.

In connection with the work of signalmen, the distance at which they are obliged to live from their signal-cabin must always be considered, for, although, no doubt, a brisk walk of a couple of miles when going to and from work, on a fine day, would rather benefit than otherwise a man who is employed for hours at a sedentary occupation, yet it must be remembered that this walk has to be performed in all weathers dry or wet, and often in snow, and signalmen should therefore, for this and other reasons, be housed wherever it is possible, within easy reach of their work.

I have, &c.,

F. A. MARINDIN,
Major, R.E.

The Assistant Secretary,
Railway Department, Board of Trade.

APPENDIX.

LIST of persons killed (10).

Mr. R. M. Boyle, Augener and Company, 86 Newgate Street, E.C.

H. P. Brodie, Plym Villa, Wallwood Road, Leytonstone.

Mr. John Lea, 111, Commercial Street, Spitalfields Market, E.

Mr. James Anderson, 8, Panmure Street, Dundee.

Mr. J. Boswell Hill, 119, George Street, Edinburgh.

Captain Duncan McLeod, the "Black Watch."

Lottie Hamilton (child), 2, Gilmore Street, Alexandria, Dumbarton.

Goods guard Petch, Middlesbro'.

Mrs. McKenzie (wife of Alex. McKenzie, injured), Edinburgh.

Miss McCulloch (aunt to L. Hamilton and sister to David McCulloch), Alexandria, Dumbarton.

LIST of persons who have complained of injury up to the 20th December.

Lady Stewart.
Mr. Geo. Rhodes.
Mr. David J. Reid.
Mr. David W. McCulloch.
Miss S. Robinson.
Rose White.
E. Burdsall.
W. Sands.
Mr. Alex. McKenzie.
Provost Sutherland.
Mrs. Dadd.
Mr. Jas. Buchanan.
Driver Roland Ewart.
Fireman Edward Head.
Mr. R. A. Hodgson.
Mr. J. L. Stewart.
Mr. A. W. Taylor.
John Rennels.
H. Wm. McNamara.
Sarah Ann Owens.
Margaret McNamara.
Annie Dilton.
Sarah Attfield.
Mrs. Milne and two children.
Miss Edith Pritchard.
Rev. J. S. Forsyth.
Miss Cheesman.
Mrs. Davis.
Miss Probert.
Anna Fennell.
Sarah Jordan.
Lily Hayler.
Amelia Higgleton.
Gertrude Waymark.
Mr. John Cook.
Mr. John Kemp.
Miss Jane Robertson.
Major-Gen. F. W. Lambton.
Mr. W. J. Harvey.
Guard Bean.
Conductor Wimpress.

DAMAGES TO ROLLING STOCK.

Damage to Engine No. 178.

The following parts were bent:—All frames, leading and trailing-axles, smoke-box, door, and dome-cover. The following parts were broken:—All wheels, both buffer-beams and buffers, leading and driving-tyres, left side-rod, leading axle-boxes, reversing-gear, all springs, all foot-steps, cab-splashers, sand-boxes, chimney, boiler-mountings, and Westinghouse brake-gear.

Damage to Tender.

The following parts were bent:—All axles, end and right side bent and broken in, frames bent and broken. The following parts were broken:—All axle boxes, Westinghouse brake-gear, all springs, and all foot-steps.